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Topic 2: Kinematics - Mark scheme

Cambridge International AS & A Level Physics (9702) | Syllabus 2025–2027

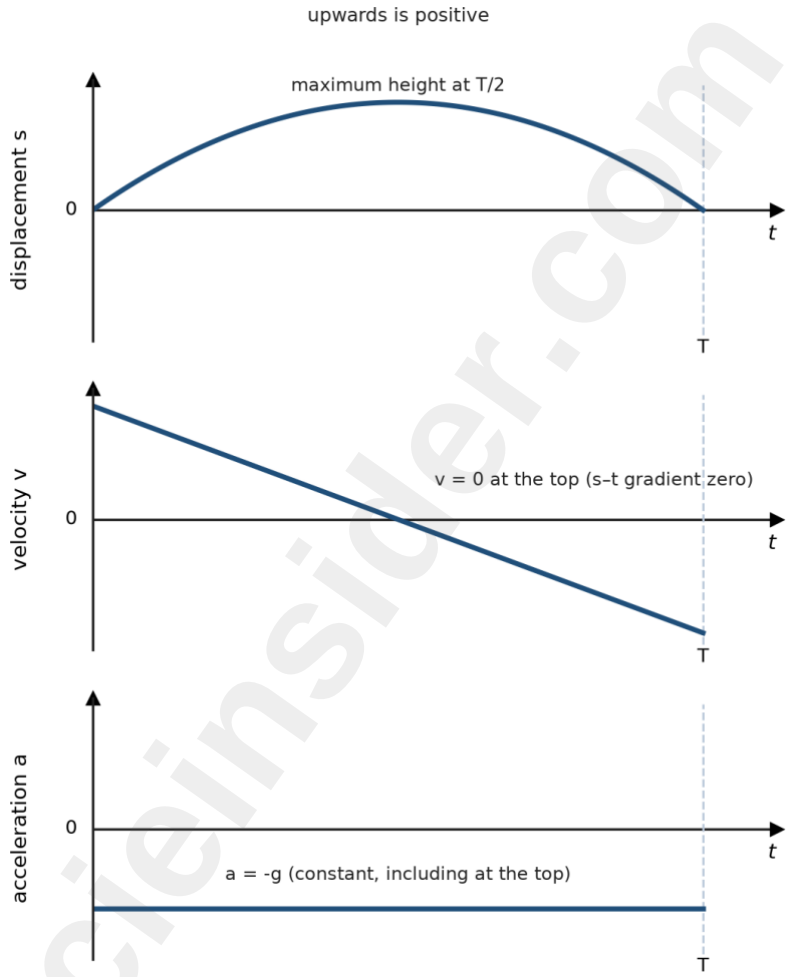
Covers: 2.1 Equations of motion

Level: AS Level

For teachers. Contains the answers to Sections 6 and 7 and model versions of the Section 4 diagrams.

Model diagrams (Section 4)

4.1 Motion graphs for a ball thrown straight up

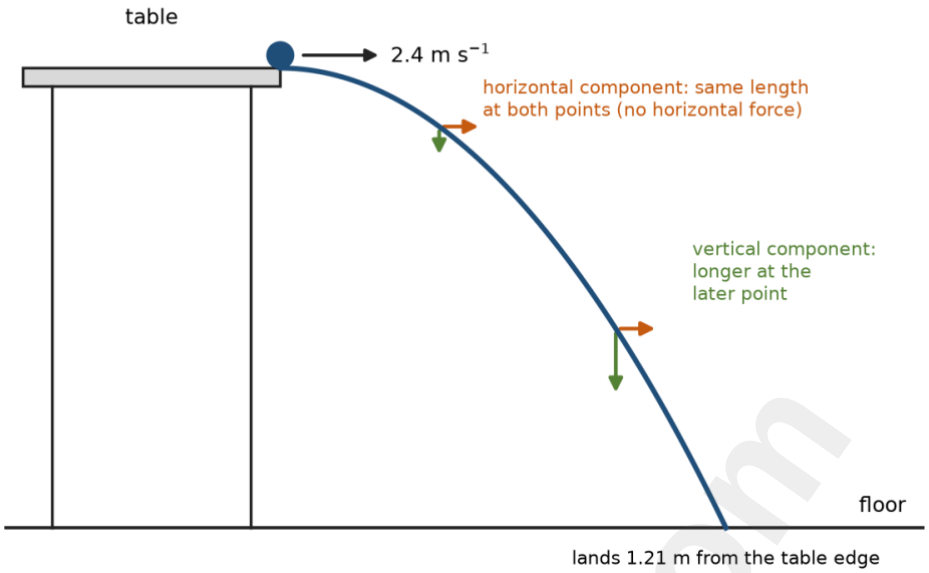


Read down the page: the gradient of the s–t curve is the velocity (zero at the top), and the gradient of the v–t line is the acceleration, −g, so the v–t line is straight and the a–t line is flat.

Credit for: s: a symmetrical curve, maximum at T/2, back to zero at T; v: a straight line starting positive, crossing zero at T/2 and ending at the same size negative; a: a horizontal line below the time axis, the same value all the way, including at the top; Nothing on the a graph reaches zero at the top: only v is zero there.

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4.2 Path of a ball rolling off a table



There is no horizontal force, so the horizontal component stays at 2.4 m s^{-1}. The vertical component grows by 9.81 m s^{-1} every second. The ball lands 1.21 m from the table edge (Question 9).

Credit for: Path starts horizontally at the table edge; Path is a smooth parabola all the way to the floor, with no vertical final section; Horizontal component arrows the same length at both points; Vertical component arrow longer at the later point; Arrows labelled horizontal and vertical.

4.3 The four equations of motion

Equation	Quantity it does not contain	Printed on the formulae page?
$v = u + at$	s (displacement)	No: recall it
$s = ½(u + v)t$	a (acceleration)	No: recall it
$s = ut + ½at^2$	v (final velocity)	Yes
$v^2 = u^2 + 2as$	t (time)	Yes

Multiple-choice answers (Section 6)

Each wrong option is a specific mistake. Where an examiner report records candidates making it, the series and paper are given in brackets. Use the third column to diagnose, not just to mark.

#	Answer	Why each wrong option is wrong
1	**A**	**B** Uses speed instead of velocity. Acceleration is a vector, so its definition needs velocity. (June 2025, Paper 21) **C** 'Rate of change' already means 'per unit time', so this is acceleration divided by time. (March 2024, Paper 12) **D** Leaves out 'change in'. Velocity divided by time is not acceleration. (March 2024, Paper 22)

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#	Answer	Why each wrong option is wrong
2	**C**	**A** Subtracts the two distances (400 − 300 = 100 m). Displacements at right angles combine by Pythagoras: 500 m. **B** Uses only the first part of the journey (400 m ÷ 100 s). **D** Works out the average speed (total distance 700 m ÷ 100 s), not the average velocity (displacement 500 m ÷ 100 s). (Nov 2022, Paper 12)
3	**D**	**A** Works out the gradient of the velocity–time graph (the deceleration, 5.0 m s^{-2}), not the area under it. **B** Multiplies the final velocity by the time: only the rectangle under the 5.0 m s^{-1} line. **C** Works out only the triangle (½ × 4.0 × 20). The area under the line is a trapezium. (March 2021, Paper 22)
4	**B**	**A** This is the graph of the vertical component of the velocity, not the acceleration. (June 2024, Paper 11) **C** Zero is the horizontal component of the acceleration. Vertically, the ball accelerates at g the whole time. **D** Gives g the wrong sign. With upwards positive, the acceleration is −9.81 m s^{-2}. (June 2024, Paper 21)
5	**B**	**A** Treats 12 m s^{-1} as an initial vertical velocity. The stone is thrown horizontally, so its initial vertical velocity is zero. **C** Divides the height by the horizontal speed (45 ÷ 12), combining the horizontal speed with the vertical distance. (Nov 2024, Paper 23) **D** Divides the height by g (45 ÷ 9.81). Use $s = \frac{1}{2}gt^2$ with an initial vertical velocity of zero.
6	**A**	**B** Has the right size but drops the sign. After 1.53 s the ball is falling, so its velocity is negative. **C** Works out gt only (9.81 × 2.00) and leaves out the initial velocity. **D** Uses a = +9.81 m s^{-2} although upwards is positive, so the signs of u and g are inconsistent. (June 2025, Paper 21)
7	**C**	**A** Only the vertical component is zero at the highest point. The horizontal component, 20 cos 30°, is unchanged. **B** Uses sin 30° instead of cos 30°: this is the initial vertical component. (June 2021, Paper 11) **D** Does not resolve the initial velocity into a horizontal component. (June 2024, Paper 21)
8	**D**	**A** $h = \frac{1}{2}gt^2$, so a graph of h against t is a curve. **B** Also a curve: the same graph with the axes swapped. **C** h is proportional to t^2, not to $1 / t^2$.

Mark scheme (Section 7)

Question 1 [3]

(a) Change in displacement per unit time / rate of change of displacement **[1]**

Not 'rate of change of displacement per unit time' (a rate of a rate). Not distance in place of displacement.

(b) Change in velocity per unit time / rate of change of velocity **[1]**

Not 'velocity / time', 'change in velocity by time' or 'rate of change of velocity per unit time'. Not speed in place of velocity.

(c) Displacement has direction (vector); distance has magnitude only (scalar). Or: displacement is measured in a straight line from the start; distance is the length of the path. **[1]**

Question 2 [8]

(a) $a = \Delta v / \Delta t = (8.0 - 2.0) / 4.0$ **[1]**

$a = 1.5$ m s^{-2} **[1]**

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(b) Area under the graph. 0 to 4.0 s (trapezium): ½ × (2.0 + 8.0) × 4.0 = 20 m **[1]**
4.0 to 7.0 s: 8.0 × 3.0 = 24 m; 7.0 to 9.0 s: ½ × 8.0 × 2.0 = 8.0 m **[1]**
Total displacement = 52 m **[1]**
A triangle for 0 to 4.0 s (½ × 4.0 × 8.0 = 16 m) loses the first mark: the line starts at 2.0 m s^{-1}.
(c) a = (0 − 8.0) / (9.0 − 7.0) **[1]**
a = −4.0 m s^{-2} (accept a deceleration of 4.0 m s^{-2}) **[1]**
(d) 52 / 9.0 = 5.8 m s^{-1} (allow error carried forward from (b)) **[1]**

Question 3 [6]

(a) v = gradient = 12 / 3.0 = 4.0 m s^{-1} **[1]**
(b) v = (0 − 12) / (8.0 − 5.0) = −4.0 m s^{-1} **[1]**
Negative: the car is moving back towards its starting point / in the opposite direction **[1]**
(c) Distance = 24 m **[1]**
Displacement = 0 **[1]**
(d) Average speed = 24 / 8.0 = 3.0 m s^{-1} **[1]**

Question 4 [5]

(a) Sketch: straight line from u at t = 0 to v at time t, with the area under it shown as a rectangle (height u) plus a triangle **[1]**
Area = ut + ½(v − u)t **[1]**
v − u = at, so s = ut + ½at^2 **[1]**
(b) From v = u + at, t = (v − u) / a **[1]**
s = ½(u + v)(v − u) / a = (v^2 − u^2) / 2a, so v^2 = u^2 + 2as **[1]**

Question 5 [6]

(a) v^2 = u^2 + 2as: 0 = 27^2 + 2 × a × 54 **[1]**
a = −6.75 m s^{-2}, so the deceleration is 6.8 m s^{-2} (accept 6.75) **[1]**
(b) t = (v − u) / a = (0 − 27) / (−6.75) **[1]**
t = 4.0 s (or s = ½(u + v)t gives 54 = ½ × 27 × t) **[1]**
(c) Thinking distance = 27 × 0.80 = 21.6 m **[1]**
Total = 21.6 + 54 = 76 m (accept 75.6 m) **[1]**

Question 6 [8]

(a) At the top v = 0: 0 = 12.0^2 − 2 × 9.81 × s **[1]**
s = 144 / 19.62 = 7.34 m **[1]**
(b) Upwards positive: u = +12.0 m s^{-1}, a = −9.81 m s^{-2}, s = −20.0 m **[1]**
−20.0 = 12.0t − 4.905t^2, so 4.905t^2 − 12.0t − 20.0 = 0 **[1]**
t = 3.58 s **[1]**
If the signs of s and g are inconsistent, the quadratic has no real solution: a useful check.
(c) v^2 = 12.0^2 + 2 × (−9.81) × (−20.0) = 536.4 **[1]**
speed = 23.2 m s^{-1} (v = u + at with t from (b) gives 23.1 to 23.2: accept) **[1]**
(d) Wrong: the acceleration is 9.81 m s^{-2} downwards throughout. Only the velocity is zero at the top. **[1]**

Question 7 [5]

Any five of the six points below, max 5.
Small, dense ball (steel) released from rest, e.g. from an electromagnet, so that air resistance is negligible **[1]**
Measure the height h fallen with a metre rule, from the bottom of the ball to the trapdoor / light gate **[1]**
Measure the time t of the fall with an electronic timer: started when the ball is released, stopped by a trapdoor or light gate **[1]**

Repeat each time measurement and average; take readings for at least five different heights **[1]**
Plot h against t^2: a straight line through the origin **[1]**
$h = \frac{1}{2}gt^2$, so g = 2 × gradient **[1]**
A hand-held stopwatch alone does not gain the third point: reaction time is large compared with a fall of about 1 m.

Question 8 [8]

(a) Horizontal: 16.0 cos 40° = 12.3 m s^{-1} **[1]**
Vertical: 16.0 sin 40° = 10.3 m s^{-1} **[1]**
(b) t = 10.28 / 9.81 = 1.05 s **[1]**
(c) $v^2 = u^2 + 2as$ with v = 0: $h = 10.28^2 / (2 \times 9.81)$ **[1]**
h = 5.39 m **[1]**
(d) Time of flight = 2 × 1.048 = 2.10 s **[1]**
Distance = 12.26 × 2.097 = 25.7 m **[1]**
(e) Horizontal: zero; vertical: 9.81 m s^{-2} downwards (both needed) **[1]**

Question 9 [6]

(a) Initial vertical velocity is zero: $1.25 = \frac{1}{2} \times 9.81 \times t^2$ **[1]**
t = 0.505 s **[1]**
(b) Distance = 2.4 × 0.505 = 1.21 m **[1]**
(c) Vertical component = 9.81 × 0.505 = 4.95 m s^{-1} (or $v^2 = 2 \times 9.81 \times 1.25$) **[1]**
speed = $\sqrt{(2.4^2 + 4.95^2)}$ **[1]**
speed = 5.50 m s^{-1} **[1]**
Adding the components (2.4 + 4.95 = 7.4 m s^{-1}) gains no credit for the last two marks: they are at right angles.

Syllabus coverage

Where each checklist statement is practised in this booklet.

Ref	Statement	Practised in
2.1.1	define and use distance, displacement, speed, velocity and acceleration	Q1, Q3(c), (d), MCQ1, MCQ2
2.1.2	use graphical methods to represent distance, displacement, speed, velocity and acceleration	Diagram 4.1, Q2, Q3, MCQ4
2.1.3	determine displacement from the area under a velocity–time graph	Q2(b), Q4(a), MCQ3
2.1.4	determine velocity using the gradient of a displacement–time graph	Q3(a), (b), Diagram 4.1
2.1.5	determine acceleration using the gradient of a velocity–time graph	Q2(a), (c), Diagram 4.1
2.1.6	derive, from the definitions of velocity and acceleration, equations that represent uniformly accelerated motion in a straight line	WE1, Q4, Diagram 4.3
2.1.7	solve problems using equations that represent uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance	WE2, Q5, Q6, MCQ5, MCQ6
2.1.8	describe an experiment to determine the acceleration of free fall using a falling object	Q7, MCQ8
2.1.9	describe and explain motion due to a uniform velocity in one direction and a uniform acceleration in a perpendicular direction	Diagram 4.2, WE3, Q8, Q9, MCQ4, MCQ7

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